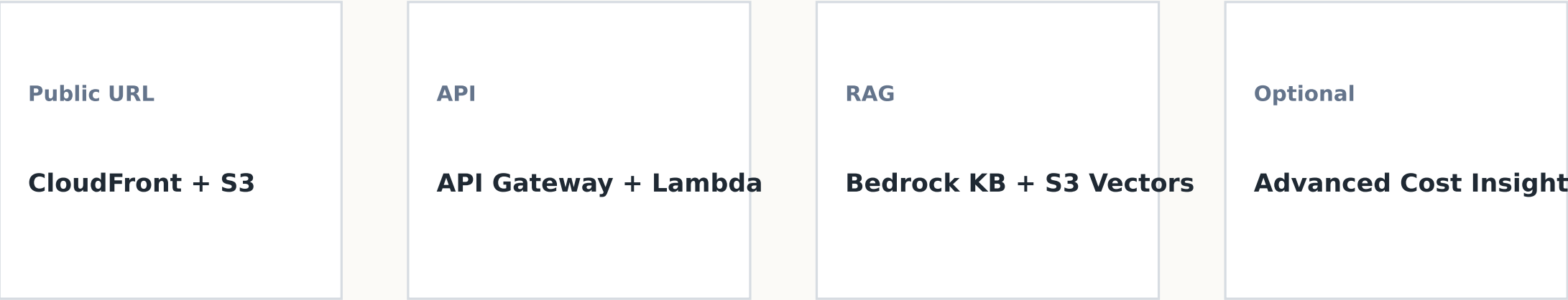


StudyBot turns static learning documents into active study workflows

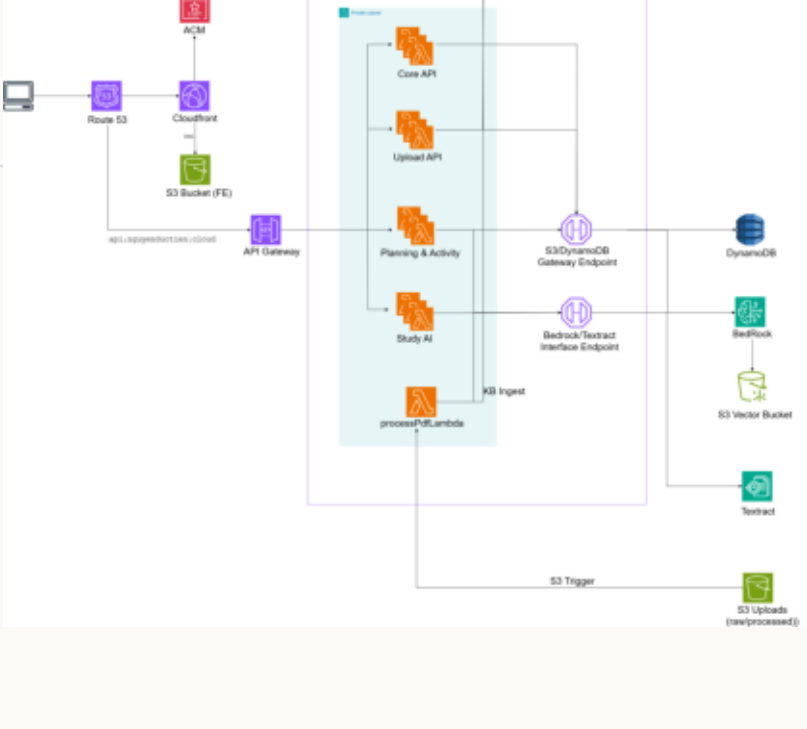
<https://nguyenductionien.cloud>

Demo: demo@studybot.com | User/session: demo / default

Ready document: w7-demo-photosynthesis



Serverless architecture keeps the demo live without idle compute



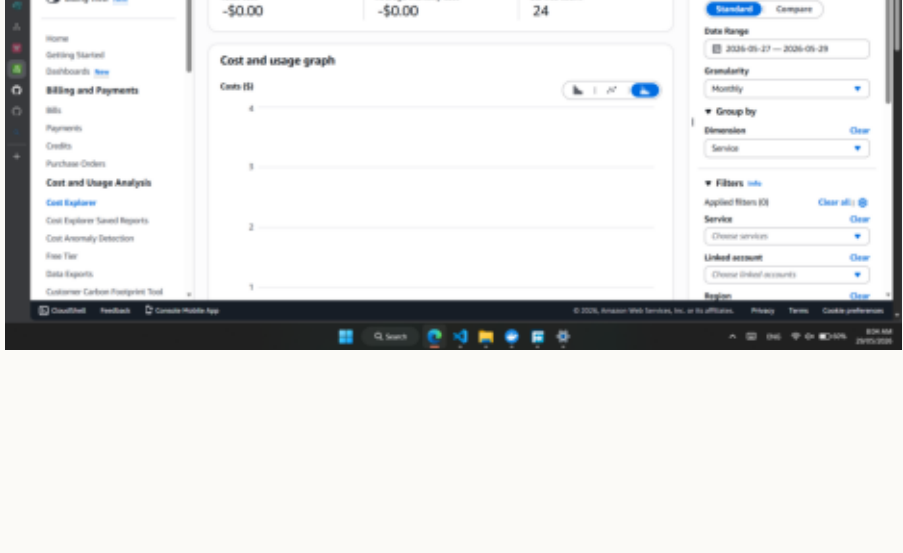
- CloudFront and private S3 host the React frontend at nguyenductionien.cloud.
- API Gateway routes login, upload, docs, Q&A, summary, quiz, planner, and dashboard to Lambda.
- DynamoDB stores users, sessions, documents, history, quizzes, summaries, and planner records.
- Bedrock Knowledge Base retrieves processed document context from S3 Vectors.

Final smoke test proves the demo path end to end

- Login succeeded for demo@studybot.com, mapped to user demo.
- docs/list returned w7-demo-photosynthesis with kb_status=READY.
- Q&A returned a grounded photosynthesis answer with citations.
- Summary, quiz, planner clarification, planner creation, dashboard, and frontend HTTP 200 all passed.

Live validation: API + persisted state + Knowledge Base ingestion were verified after deploy.

Advanced Cost Insights is the selected optional capability



- Budget: StudyBot-W7-weekly-safety-net, USD 100 limit, May 27-June 4.

- Cost Anomaly Detection: Default-Services-Monitor with confirmed daily email subscription.

- Cost Explorer screenshot covers May 27-29 Friday pre-demo service spend view.

- Cost-per-feature table maps upload/ingestion, Q&A, summary, quiz, planner, and frontend serving.

Tagging strategy separates ownership, cost review, and teardown discovery

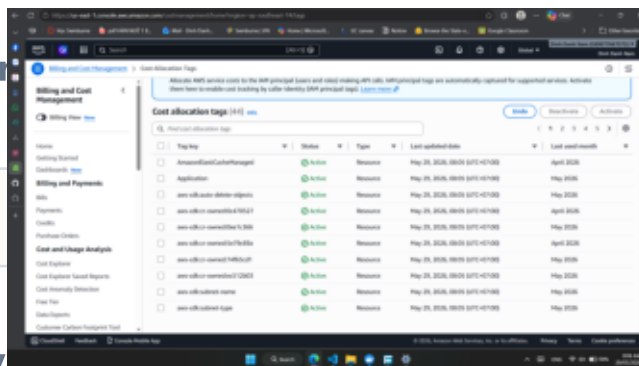
Project **W7Capstone**

- Tags are applied at CDK stack level so redeployable resources inherit the same metadata.

Team **G11**

- Resource Groups Tagging API found 49 resources tagged with Project=W7Capstone and Team=G11.

Ow



Environment & Infrastructure Protection

- Cost allocation tags are activated, but Cost Explorer tag filtering can lag and is not retroactive.

Retrieval was tuned for short study paragraphs and demo stability

- Generation model: Nova 2 Lite, chosen for long context, large output, and Bedrock-native integration.
- Vector store: S3 Vectors with Bedrock Knowledge Base, chosen over OpenSearch/Aurora to reduce fixed cost.
- Chunking: fixed 300 tokens with 15% overlap; sample paragraphs and VTT cues were mostly short.
- Embedding model: cohere.embed-multilingual-v3 retained for Vietnamese/English support and 1024-dimension index compatibility.



Security and monitoring evidence support the live system

- IAM roles are scoped by feature boundary across login, sessions, upload, documents, Q&A, summary, quiz, planner, history, and ingestion.

- Lambdas run in private isolated subnets and use VPC endpoints for AWS service access.

- CloudWatch dashboard, alarms, and Logs Insights track API 5XX, Lambda errors, p95 duration, and ingestion failure signals.

- Observability is supporting evidence; Advanced Cost Insights is the claimed optional capability.

Teardown is prepared but intentionally waits until grading is complete

- cdk destroy StudyBotInfraStack is documented and targets the W7 stack.
- Post-demo proof to capture: CloudFormation deletion status, Resource Groups Tagging API residual check, and final Cost Explorer proof.
- Teardown is not run before grading because it would remove the live URL and API needed for assessment.
- docs/teardown_confirmation.md contains the checklist and resource IDs.